

Olympiad Test Paper — Maths (Class 7) — Paper 4

Chapter 13: Connecting the Dots

Paper 4 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	13 — Connecting the Dots
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: averaging two means without weighting by group size, forgetting to re-sort after editing a value, confusing the change in the mean with the new data point itself, treating an unchanged median as proof nothing moved, and trusting a bar's apparent height across a truncated axis or a switched scale. Do not write on this paper — mark answers on your answer sheet.

1. The mean of six numbers is 40. Two of the numbers are 30 and 50. If those two are removed, the mean of the remaining four numbers is:

- (A) 38 (B) 42
(C) 55 (D) 40

2. The mean of 8 numbers is 15. A 9th number is added and the mean becomes 16. A 10th number is then added and the mean returns to 15. The 10th number is:

- (A) 15 (B) 24
(C) 6 (D) 16

3. Every number in a data set of 5 values is first doubled and then 3 is subtracted from each result. If the original mean was 10, the new mean is:

- (A) 20 (B) 17
(C) 10 (D) 23

4. Class A (mean 60) and Class B (mean 80) are combined and the overall mean is exactly 68. What is the ratio of the number of students in Class A to Class B?

- (B) 3 : 2
- (D) 3 : 1

5. Seven numbers in increasing order are 6, 9, 13, x, 22, 27, 31. It is required that the median equals one quarter of the range. The value of x:

- (A) 6.25 (B) 25
- (C) does not exist, because the median is the 4th value x (which must be between 13 and 22), but $\text{range} \div 4 = 25 \div 4 = 6.25$ is too small to sit there
- (D) 18

6. A list of five distinct whole numbers has median 12. The two largest are each increased by 10 and the smallest is decreased by 4 (the order is preserved). What is the new median?

- (A) 22 (B) 8
(C) 12 (D) 14

7. A data set is 3, 3, 5, 7, 7, 7, 9. The fewest single-entry edits (each changes one number) needed to make the data bimodal with modes 3 and 7 is:

- (A) 0 — it is already bimodal (B) 2
(C) 3
(D) 1 (change the 9 into a 3, so 3 appears three times like 7)

8. Two cricketers each played 5 innings, both with mean 40. Asha's scores have range 30; Bina's range 12. A coach says 'equal means prove equal skill.' The sound comparison is:

- (A) Asha is better because a larger range means a higher ceiling
(B) They are identical players since the means are equal
(C) Bina is worse because a small range means low scores
(D) Their totals are equal (both 200), but Bina is more consistent because her smaller range shows her scores cluster more tightly

9. The mean of five numbers is 18. When the largest number is removed, the mean of the remaining four falls to 15. The number that was removed is:

- (A) 18
(B) 15
(C) 3
(D) 30

10. A data set of 7 values is 4, 6, 8, 10, 12, 14, 16 (mean 10, median 10). The value 16 is replaced by 100. Which statement is exactly correct?

- (A) The mean rises to 22 while the median stays at 10
(B) Both the mean and the median rise to 22
(C) The mean stays 10 but the median rises
(D) Neither changes because only one value moved

11. The mean of 10 numbers is 12. Two numbers, 5 and 9, are removed and replaced by a single number 22. The new mean (of the now 9 numbers) is:

- (A) 12 (B) Approximately 14.2
(C) 12.8 (D) 22

- 12.** The mean of 12 observations was computed as 30, but later TWO entries were found wrong: 18 had been recorded as 8, and 25 had been recorded as 52. The corrected mean is:
- (A) 30 (B) Approximately 28.6
(C) 31.4 (D) 32.5
- 13.** A double bar graph shows a shop's sales across 2023 and 2024. Tea: 2023 $\rightarrow$ 80, 2024 $\rightarrow$ 110. Coffee: 2023 $\rightarrow$ 140, 2024 $\rightarrow$ 119. Which single statement is fully correct?
- (A) Both products grew because both 2024 bars are tall
(B) Tea overtook Coffee in 2024
(C) Coffee grew faster than Tea because its bars are taller overall
(D) Tea grew by 30 while Coffee fell by 21, yet Coffee's 2024 sales (119) still exceed Tea's (110)
- 14.** Adding 7 to each number of a data set produces a new mean of 19. The original total of the data was 96. The number of values, n , is:
- (A) 8 (B) 12
(C) 5 (D) 7
- 15.** A pictograph uses one symbol = 12 books, with quarter-symbols allowed. Shelf P shows $3\frac{3}{4}$ symbols and Shelf Q shows $2\frac{1}{4}$ symbols. How many MORE books are on Shelf P than Shelf Q?
- (A) 1.5 (B) 18
(C) 12 (D) 72
- 16.** Three groups have means 10, 20 and 30. The first group has 2 numbers and the third has 3 numbers. If the overall mean of all the numbers is 21, the second group contains how many numbers?
- (A) 5 (B) 1
(C) 10 (D) 3
- 17.** A magazine's bar graph has a vertical axis starting at 200. Region X's bar (value 212) appears about four times as tall as Region Y's bar (value 203). The honest conclusion about the ratio of actual values is:
- (A) X is only about 1.04 times Y; the truncated axis turned a 9-unit gap into a fourfold visual illusion
(B) X is about four times Y, as the bars show
(C) Y is essentially zero
(D) The graph cannot be read at all
- 18.** Eight numbers in increasing order are 4, 7, 11, a , b , 20, 24, 29 with $a < b$. Their median is 16. Which pair (a , b) is IMPOSSIBLE?
- (A) $a = 14$, $b = 18$ (B) $a = 15$, $b = 17$
(C) $a = 17$, $b = 15$ (D) $a = 13$, $b = 19$

19. Seven numbers written in increasing order are 5, 8, x , 14, 18, 18, 25. The data has mean 14 and a unique mode of 18. The value of x is:

- (A) 10
- (B) 14
- (C) 12
- (D) 9

20. In a data set every value appears once except one value v that appears three times. Two copies of a new value w ($w \neq v$, previously absent) are added. After this, the mode is:

- (A) w , the unique mode
- (B) still v alone — w reaches only two copies, fewer than v 's three, so the set stays unimodal
- (C) both v and w , a tie
- (D) no mode at all

21. Five positive whole numbers have mean 8, median 7, and a unique mode of 5. Which set satisfies all three conditions?

- (A) 5, 5, 8, 8, 14
- (B) 4, 5, 5, 7, 19
- (C) 5, 5, 7, 9, 13
- (D) 5, 5, 7, 8, 15

22. A bar graph (scale 1 cm = 6 marks) shows four students' bars of 5 cm, 7 cm, 6 cm and 6 cm. A fifth student scored exactly the mean of those four. The fifth student's bar height would be:

- (A) 6 cm
- (B) 36 cm
- (C) 24 cm
- (D) 1 cm

23. The mean of 9 numbers is 20. Three of the numbers, each equal to 20, are removed. The mean of the remaining six numbers is:

- (A) 20
- (B) 13.3
- (C) 30
- (D) 60

24. A list of five distinct numbers has range 20. The smallest is reduced by 5 and the largest is increased by 5 (order preserved). The new range is:

- (A) 20
- (B) 30
- (C) 25
- (D) 15

25. A data set of 6 numbers has mean 50. A 7th value equal to 50 and an 8th value equal to 90 are added. The final mean of all 8 numbers is:

- (A) 50
- (B) 63.3
- (C) 55
- (D) 70

26. Class A has 30 students with mean 78. After merging with Class B (mean 66), the combined mean is 72. How many students are in Class B?

- (A) 20
- (B) 30
- (C) 36
- (D) 60

27. The mean of a data set is 16. Each value is replaced by $(3 \times \text{the value} - 8)$. The mean of the new set is:

(A) 48

(B) 16

(C) 40

(D) 24

28. A list of seven distinct numbers has median 25. The median value 25 is itself removed, leaving six numbers. The new median is:

(A) still 25

(B) the average of the numbers that had been immediately on either side of 25 (the original 3rd and 5th values)

(C) the original 3rd value

(D) the original 5th value

29. A set of 10 numbers has mode 6 appearing exactly 4 times and value 9 appearing once. Three copies of 9 and one extra copy of 6 are added. After the additions, the mode is:

(A) 9, appearing 4 times

(B) 6, appearing 5 times (more than 9's total of 4)

(C) both 6 and 9, a tie

(D) there is no mode

30. The mean of 6 numbers is 21. Replacing the smallest number (which is 9) by an unknown value w makes the new mean 24. The value of w is:

(A) 27

(B) 24

(C) 18

(D) 33

31. A magazine claims 'Sales doubled' and shows a bar graph whose vertical axis starts at 480. The first bar reaches 490 and the second reaches 500, and the second bar's visible part is indeed twice the first's. The honest description is:

(A) Sales genuinely doubled, from 490 to 500 (B) Sales fell, since the bars are short

(C) The values cannot be determined

(D) Sales rose only from 490 to 500 (about 2%); the axis starting at 480 made a 10-unit gain look like a doubling

32. The mean of a list is 30. A value equal to 30 is removed, and two new values, 10 and 50, are added. The list originally had 8 numbers. The new mean is:

(A) 33.3

(B) 30

(C) 28

(D) 37.5

33. The mean of 10 numbers was computed as 50. Later it was found that a value of 84 had been entered as 48, and a value of 23 had been entered as 32. The corrected mean is:

(A) 50

(B) Approximately 52.7

(C) 54.5

(D) 47.3

34. A data set of 7 numbers has mean 24 and median 24. The smallest value (originally 10) is decreased to 2, and the largest (originally 40) is increased to 60, order preserved. Which statement is exactly true?

- (A) The median stays 24 while the mean rises (the net change +12 lifts the mean to about 25.7)
- (B) Both the mean and the median rise (C) Both the mean and the median stay 24
- (D) The median rises but the mean stays the same

35. The mean of three numbers is 14 and the mean of four other numbers is 21. A single extra number x is included with all seven; the grand mean of all eight numbers becomes 20. The value of x is:

- (A) 20 (B) 34
- (C) 18 (D) 126

36. A data set of 6 numbers has mean 10 and median 9. The largest value, originally 18, is increased to 30. After this change, which is guaranteed?

- (A) The mean increases by 2 (to 12) while the median stays 9
- (B) The mean and median both increase (C) The mean stays 10, the median increases
- (D) Neither the mean nor the median changes

37. A double bar graph compares boys and girls in two clubs. Music: boys 15, girls 9. Art: boys 6, girls 22. Which single statement is fully correct?

- (A) Music is more popular overall and has the bigger gap
- (B) Art is more popular but Music has the bigger gap
- (C) Art's total (28) exceeds Music's (24), so Art is more popular; and Art's gender gap $|6 - 22| = 16$ exceeds Music's gap $|15 - 9| = 6$
- (D) Both clubs have equal totals so neither is more popular

38. A pictograph uses one symbol = 20 trees and a bar graph elsewhere uses scale 1 cm = 8 trees. The pictograph shows $2\frac{1}{2}$ symbols for Village P; the bar graph shows a 9 cm bar for Village Q. Which village planted more, and by how many trees?

- (A) Village Q, by 22 trees (B) Village P, by 22 trees
- (C) they planted the same number (D) Village Q, by 6.5 trees

39. A data set of 5 numbers has mean 20 and median 20. The two largest values are each increased by 6 and the two smallest are each decreased by 6. Which is exactly true after the change?

- (A) The mean rises by 6 and the median stays 20
- (B) The mean stays 20 but the median rises (C) Both fall by 6
- (D) The mean stays 20 and the median stays 20

40. The mean of 5 numbers is 12. Each number is increased by its own position number (1st by 1, 2nd by 2, ..., 5th by 5). The new mean is:

- (A) 12 (B) 17
- (C) 15 (D) 13

41. A list of 9 numbers has range 50. The largest value is decreased by 12 and the smallest is increased by 8 (order otherwise unchanged). The new range is:

- (A) 50
- (B) 70
- (C) 30
- (D) 20

42. A bar graph uses scale 1 cm = 6 units, but a printing error stretched every bar's HEIGHT (not its value) by a factor of 1.5. A bar that should represent 30 units now physically measures:

- (A) 5 cm
- (B) 7.5 cm
- (C) 45 cm
- (D) 30 cm

43. Six numbers have mean 15 and median 14. The two smallest values, originally 6 and 8, are both replaced by 2. After the change, which is exactly true?

- (A) The mean stays 15 and the median stays 14
- (B) Both the mean and the median fall by 2
- (C) The mean falls to 13.4 and the median also falls to 12
- (D) The mean falls while the median stays 14

44. A set of 11 numbers has mode 7 appearing 5 times. Two copies of 4 (which appeared 3 times before) are added. After the additions, the mode is:

- (A) 7 alone, with 5 copies
- (B) 4 alone, with 5 copies
- (C) both 7 and 4 — each appears 5 times, so the data is bimodal
- (D) there is no mode

45. Group P (mean 30) has twice as many numbers as Group Q (mean 60). The mean of all the numbers combined is:

- (A) 45
- (B) 50
- (C) 40
- (D) 90

46. A data set of 5 numbers has mean 14 and range 24. The two extreme values are 4 and 28. A coach says 'the spread of 24 means the typical value is also far from the mean.' The mathematically correct comment is:

- (A) A range of 24 proves every value is at least 12 away from the mean
- (B) The range equals the mean, so the data is balanced
- (C) A large range always means the median is far from the mean
- (D) Range only measures the gap between the two extremes; it says nothing on its own about how the middle three values sit, which could still cluster near 14

47. A bar graph (scale 1 unit = 5 km) shows three trips of 4, 6 and 8 units. A bus's odometer says the three trips totalled 95 km. By how many km does the graph total fall short of the odometer?

- (A) 5 km
- (B) 18 km
- (C) 0 km
- (D) 90 km

- 48.** The marks of nine students are 8, 12, 12, 14, 14, 14, 16, 16, 20. Compute (mean + mode – 2 × median).
- (A) 2 (B) –2
(C) 0 (D) 14
- 49.** The marks of seven students are 12, 15, 15, 18, 21, 21, 21. The correct increasing order of mean, median and mode is:
- (A) mean < median < mode (B) median < mean < mode
(C) mode < median < mean (D) mean = median = mode
- 50.** The mean of a data set is 25. Each value is first halved, then 10 is added to each result. The mean of the new set is:
- (A) 22.5 (B) 12.5
(C) 35 (D) 17.5
- 51.** The mean of 12 numbers was found to be 40, but a single value was entered as 19 when it should have been 91. The corrected mean is:
- (A) 40 (B) 43
(C) 52 (D) 46
- 52.** Nine numbers in increasing order have median 40 (the 5th value). The four LARGEST values (positions 6 to 9) are each increased by 100, and the four SMALLEST (positions 1 to 4) are each decreased by 100. The new median is:
- (A) 140 (B) –60
(C) 20 (D) 40 (unchanged)
- 53.** A cricketer's mean over 10 innings is 45. He wants to raise his mean to 50 over the next innings (his 11th). How many runs must he score in that one innings?
- (A) 100 (B) 50
(C) 55 (D) 5
- 54.** A data set of 6 numbers has mean 20. Three of the numbers are each equal to the mean. The other three have a known sum. If the other three sum to S, then S equals:
- (A) 120 (B) 60
(C) 20 (D) 40
- 55.** Two pictographs use DIFFERENT scales: in graph 1, one symbol = 15 apples; in graph 2, one symbol = 10 apples. Basket A shows 4 symbols on graph 1; Basket B shows 5 symbols on graph 2. Which basket has more apples, and by how many?
- (A) Basket B, by 1 apple (B) Basket A, by 1 apple
(C) they are equal (D) Basket A, by 10 apples
- 56.** The mean of five numbers is 30. The smallest is 10 and the largest is 60. The middle three numbers are all equal to each other. Each of those middle numbers equals:

(A) 30

(B) 40

(C) 80

(D) About 26.7

57. A double bar graph shows rainfall (mm) for two cities over three months. City P: 60, 90, 30 (total 180). City Q: 50, 50, 80 (total 180). A reporter writes 'identical rainfall.' The most accurate correction is:

(A) The totals differ, so the reporter is right to call them different

(B) They are identical because the totals are equal

(C) Their totals match (180 each), but the monthly pattern differs — P peaks in month 2 while Q peaks in month 3 — so the rainfall is not identical month by month

(D) City Q had more rain because its highest single month (80) is the largest value

58. The mean of 7 numbers is 18. If the largest number, 30, is replaced by a value that makes the mean a whole number 16, the replacement value is:

(A) 14

(B) 2

(C) 30

(D) 16

59. A set of data has mean 40, median 40 and mode 40. A new value of 100 is added. Which statement is guaranteed correct?

(A) All three rise to about the same new value

(B) The mode jumps to 100 because 100 is the new largest value

(C) The mean rises above 40, while the mode stays 40 and the median rises by at most a little (it does not jump to 100)

(D) Nothing changes because the data was already balanced

60. The mean of 6 numbers is 12. A 7th and an 8th number are added; the 7th is double the 8th, and the mean of all 8 becomes 15. The 8th number is:

(A) 24

(B) 48

(C) 16

(D) 8

Solutions — Maths (Class 7), Chapter 13:

Connecting the Dots — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	D	31	D	41	C	51	D
2	C	12	B	22	A	32	B	42	B	52	D
3	B	13	D	23	A	33	B	43	D	53	A
4	B	14	A	24	B	34	A	44	C	54	B
5	C	15	B	25	C	35	B	45	C	55	D
6	C	16	A	26	B	36	A	46	D	56	D
7	D	17	A	27	C	37	C	47	A	57	C
8	D	18	C	28	B	38	A	48	C	58	D
9	D	19	A	29	B	39	D	49	A	59	C
10	A	20	B	30	A	40	C	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Total of all six = $6 \times 40 = 240$. The two removed numbers sum to $30 + 50 = 80$, so the remaining four total $240 - 80 = 160$, and their mean is $160 \div 4 = 40$. The two removed values average 40 (the same as the overall mean), so removing them leaves the mean unchanged at 40. Dividing the remaining total by 6 or 5 instead of 4 gives the off-values near 38 or 42; 55 wrongly works with the removed sum 80 instead of the remaining total.

2. (C) Eight numbers total $8 \times 15 = 120$. After the 9th, the total is $9 \times 16 = 144$, so the 9th number = $144 - 120 = 24$. After the 10th, the total must be $10 \times 15 = 150$, so the 10th number = $150 - 144 = 6$. The value 24 is the 9th number, not the 10th; 15 just copies the final mean; 16 copies the intermediate mean. Only 6 brings the average back to 15.

3. (B) A transform of the form new = $2 \times \text{old} - 3$ applies the same way to the mean: new mean = $2 \times 10 - 3 = 20 - 3 = 17$. Stopping after doubling gives 20; leaving the mean unchanged ignores the transform; doubling then ADDING 3 (mishandling the sign) gives 23.

4. (B) Let A have a students and B have b . Then $(60a + 80b) \div (a + b) = 68$, so $60a + 80b = 68a + 68b$, giving $12b = 8a$, i.e. $a : b = 12 : 8 = 3 : 2$. Because 68 is closer to 60 than to 80, the lower-mean class A must be larger — consistent with $3 : 2$. A $1 : 1$ split gives mean 70; $2 : 3$ leans toward 80 (mean 72); $3 : 1$ gives mean 65.

5. (C) With seven sorted values the median is the 4th value, which is x , so x must satisfy $13 \leq x \leq 22$. The range $= 31 - 6 = 25$, so $\text{range} \div 4 = 6.25$. The condition $\text{median} = \text{range} \div 4$ would force $x = 6.25$, but 6.25 lies far below 13 and cannot occupy the 4th sorted slot. Hence no valid x exists. Writing $x = 6.25$ ignores the ordering constraint; 25 mistakes the range for x ; 18 is an unjustified guess that does not equal 6.25.

6. (C) For five sorted values the median is the 3rd. Only the two largest (positions 4, 5) and the smallest (position 1) were changed, leaving the 3rd value untouched, so the median stays 12. The median resists changes in the extremes. Adding 10 (giving 22) or subtracting 4 (giving 8) wrongly assumes an edited value reaches the middle; 14 invents a shift that never touches position 3.

7. (D) Currently 7 appears three times and 3 appears twice, so 7 is the unique mode — NOT already bimodal. To tie 3 with 7, make 3 appear three times: change the single 9 into a 3, giving 3, 3, 3, 5, 7, 7, 7, where both 3 and 7 appear three times. That is one edit. Believing it is already bimodal ignores that 3 has only two copies; two or three edits exceed what is needed.

8. (D) Mean 40 over 5 innings gives each a total of 200, so totals match. Range measures spread: Bina's range 12 keeps her scores in a tight band, while Asha's range 30 swings much wider. Smaller spread about the same mean signals greater consistency, so Bina is steadier. Equal means do not imply identical data; a large range is not a 'higher skill ceiling'; and a small range describes tight clustering, not low values.

9. (D) Five numbers total $5 \times 18 = 90$; four numbers total $4 \times 15 = 60$. The removed (largest) number $= 90 - 60 = 30$. Since removing it lowered the mean, it was above average, and $30 > 18$ confirms this. 18 copies the original mean; 15 copies the reduced mean; 3 is merely the change in mean ($18 - 15$).

10. (A) New total $= 70 - 16 + 100 = 154$, so the new mean $= 154 \div 7 = 22$ (up by 12). The sorted list 4, 6, 8, 10, 12, 14, 100 still has 10 as its 4th (middle) value, so the median is unchanged at 10. This is the outlier signature: the mean is dragged by an extreme value while the median resists. Claiming both reach 22 ignores the median's robustness; saying the mean stays 10 ignores the changed total; replacing a value certainly changes the mean.

11. (B) Original total $= 10 \times 12 = 120$. Removing 5 and 9 (sum 14) and adding 22 gives a new total of $120 - 14 + 22 = 128$ over 9 numbers, so the mean $= 128 \div 9 \approx 14.2$. The count dropped from 10 to 9 because two values became one. Dividing 128 by 10 gives 12.8 (wrong count); 12 copies the old mean; 22 mistakes the inserted value for the mean.

12. (B) Recorded total = $12 \times 30 = 360$. First error: 8 should be 18, so add 10. Second error: 52 should be 25, so subtract 27. Net correction = $+10 - 27 = -17$. True total = $360 - 17 = 343$, mean = $343 \div 12 \approx 28.6$. The two errors pull oppositely and the negative one dominates, so the mean falls slightly. Leaving it at 30 ignores both corrections; 31.4 applies them with the wrong sign; 32.5 accounts for only one error.

13. (D) Tea: $80 \rightarrow 110$, a rise of 30. Coffee: $140 \rightarrow 119$, a fall of 21. So Tea grew and Coffee declined. Comparing 2024 values, Coffee $119 >$ Tea 110 , so Coffee still leads despite shrinking. Tall bars do not mean growth (Coffee fell); Tea did not overtake Coffee ($110 < 119$); and Coffee did not grow at all, so it cannot have grown faster.

14. (A) Adding 7 to each value raises the mean by 7, so the original mean = $19 - 7 = 12$. Since original mean $\times n$ = original total, $12 \times n = 96$, giving $n = 8$. The key step is undoing the shift before using the total. Using 19 directly gives $96 \div 19 \approx 5$; 12 is the recovered old mean mistaken for the count; 7 is the shift, not the count.

15. (B) Shelf P = $3.75 \times 12 = 45$ books; Shelf Q = $2.25 \times 12 = 27$ books; difference = $45 - 27 = 18$. Equivalently $(3.75 - 2.25) \times 12 = 1.5 \times 12 = 18$. Subtracting symbol counts alone (1.5) forgets each symbol is 12 books; 12 is the value of one symbol; 72 is the combined total ($45 + 27$), not the difference.

16. (A) Let the second group have k numbers. Totals: group1 = $2 \times 10 = 20$; group2 = $20k$; group3 = $3 \times 30 = 90$. Overall: $(20 + 20k + 90) \div (2 + k + 3) = 21$, i.e. $(110 + 20k) \div (k + 5) = 21$, so $110 + 20k = 21k + 105$, giving $k = 5$. Check: total 210 over 10 numbers = 21 ✓. Choosing 1 or 3 fails the equation; 10 mistakes the total count for the group size.

17. (A) The axis starts at 200, so visible bar lengths measure only the parts above 200: X shows 12 units, Y shows 3 units, hence the 4-to-1 look. But the true values are 212 and 203, whose ratio is $212 \div 203 \approx 1.04$ — nearly equal, differing by just 9. The fourfold appearance is an artefact of the cut axis. X is not four times Y, Y is clearly 203 (not zero), and the graph is readable once the axis start is noted.

18. (C) For eight values the median is the average of the 4th and 5th values, a and b , so $(a + b) \div 2 = 16$, i.e. $a + b = 32$. The pairs (14, 18), (15, 17) and (13, 19) all sum to 32 and respect $a < b$ and $11 < a$, $b < 20$. The pair (17, 15) sums to 32 but violates $a < b$ ($17 > 15$), so it cannot occupy positions 4 and 5 in increasing order. The trap is checking only the sum and ignoring the ordering.

19. (A) With seven numbers, total = $7 \times 14 = 98$. The six known values sum to $5 + 8 + 14 + 18 + 18 + 25 = 88$, so $x = 98 - 88 = 10$. Check the constraints: the list becomes 5, 8, 10, 14, 18, 18, 25, which is still increasing ($8 < 10 < 14$) ✓; 18 appears twice and every other value once, so 18 is the unique mode ✓; and the mean is $98 \div 7 = 14$ ✓. The value 14 mistakes the mean for x ; 12 averages the neighbours 8 and 14 instead of using the total; 9 is an unjustified guess that makes the sum only 97, so the mean would not be 14.

20. (B) Originally v appears three times (the unique mode) and every other value once. Adding two copies of w makes w appear $0 + 2 = 2$ times, still fewer than v 's three. So v remains the single most frequent value: the set stays unimodal. To tie, w would need a third copy. Claiming w becomes the mode, that both tie, or that the mode vanishes all overstate what two copies achieve.

21. (D) Need: five numbers, total $5 \times 8 = 40$; the 3rd sorted value $= 7$; the value 5 the unique most frequent. Test 5, 5, 7, 8, 15: sum $= 40$ ✓, 3rd value 7 ✓, 5 appears twice and is the only repeat ✓. The set 5, 5, 8, 8, 14 has two values each appearing twice (5 and 8) → bimodal, so 5 is not the unique mode. The set 4, 5, 5, 7, 19 has 3rd value 5, so its median is 5, not 7. The set 5, 5, 7, 9, 13 sums to 39, not 40, so its mean is not 8. Only 5, 5, 7, 8, 15 meets every condition.

22. (A) Bar heights in cm are 5, 7, 6, 6; their mean height $= (5 + 7 + 6 + 6) \div 4 = 24 \div 4 = 6$ cm. Because the scale is linear, the bar for the mean mark also has the mean height, 6 cm. (Check: marks 30, 42, 36, 36, mean 36, and $36 \div 6 = 6$ cm.) The value 36 is the mean in marks, not cm; 24 is the total height; 1 confuses dividing height by the scale. The bar is 6 cm tall.

23. (A) Total of nine $= 9 \times 20 = 180$. Removing three values each equal to 20 removes 60, leaving 120 over six numbers, mean $= 120 \div 6 = 20$. Removing values equal to the current mean leaves the mean unchanged. $13.3 = 120 \div 9$ keeps the wrong count; 30 wrongly raises the mean; 60 is the removed total.

24. (B) Range $=$ largest $-$ smallest $= 20$. After the changes, the smallest drops by 5 and the largest rises by 5, so the new range $=$ (largest $+ 5$) $-$ (smallest $- 5$) $=$ (largest $-$ smallest) $+ 10 = 20 + 10 = 30$. Both ends move outward, widening the spread by 10. Keeping 20 ignores both moves; 25 adds only one 5; 15 subtracts instead of adding.

25. (C) Six numbers total $6 \times 50 = 300$. Adding 50 keeps the total at 350 (mean 50 over 7). Adding 90 gives total 440 over 8 numbers, mean $= 440 \div 8 = 55$. The 50 leaves the mean alone, but the 90 (above the mean) pulls it up to 55. Staying at 50 ignores the 90; 63.3 uses a wrong count; 70 naively averages 50 and 90, ignoring the six existing values.

26. (B) Let Class B have b students. $(30 \times 78 + 66b) \div (30 + b) = 72$, i.e. $(2340 + 66b) \div (30 + b) = 72$, so $2340 + 66b = 2160 + 72b$, giving $180 = 6b$, $b = 30$. Since 72 is exactly midway between 78 and 66, the groups must be equal in size, confirming 30. Values 20 and 36 break the balance; 60 over-weights B and would pull the mean below the midpoint.

27. (C) A linear transform $\text{new} = 3 \times \text{old} - 8$ applies to the mean directly: new mean $= 3 \times 16 - 8 = 48 - 8 = 40$. The multiplier scales the mean and the constant shifts it. Stopping after multiplying gives 48; leaving the mean unchanged ignores the transform; 24 mishandles the constant (e.g. $16 + 8$) instead of $3 \times 16 - 8$.

28. (B) In seven sorted values the median is the 4th value, 25. Removing it leaves the original 1st, 2nd, 3rd, 5th, 6th, 7th. The new median of six values is the average of the new 3rd and 4th, which are the original 3rd and 5th — the immediate neighbours of 25. So the new median is their average, not 25 (that value is gone) and not just one neighbour.

29. (B) Initially 6 appears 4 times and 9 once. Adding three 9s makes 9 appear $1 + 3 = 4$ times. Adding one more 6 makes 6 appear $4 + 1 = 5$ times. So 6 (5 times) beats 9 (4 times) and stays the unique mode. The extra 6 is decisive — without it there would be a tie at 4. Choosing 9 ignores the added 6; the tie claim ignores 6 reaching 5; 'no mode' is false.

30. (A) Original total = $6 \times 21 = 126$; new total = $6 \times 24 = 144$, an increase of 18. Replacing 9 by w changes the total by $(w - 9)$, so $w - 9 = 18$, giving $w = 27$. The mean rose by 3 over 6 numbers, i.e. a total rise of 18 carried entirely by this swap. 24 copies the new mean; 18 is the total change, not w ; $33 = 24 + 9$ adds the old value instead of solving $w - 9 = 18$.

31. (D) With the axis starting at 480, the visible heights are $490 - 480 = 10$ and $500 - 480 = 20$, so the second looks twice the first. But the real values are 490 and 500: a rise of just 10, about 2% — nowhere near doubling. The truncated axis created the illusion. Sales did not double, did not fall, and the values are clearly readable as 490 and 500.

32. (B) Original total = $8 \times 30 = 240$. Removing a 30 leaves 210 over 7. Adding 10 and 50 (sum 60, averaging 30 — the same as the mean) gives total 270 over 9 numbers, mean = $270 \div 9 = 30$. Removing a value equal to the mean and adding two values whose average equals the mean both leave the mean unchanged. 33.3 and 37.5 use wrong counts; 28 wrongly assumes the additions drag the mean down.

33. (B) Recorded total = $10 \times 50 = 500$. First error: 48 should be 84, add 36. Second error: 32 should be 23, subtract 9. Net correction = $+36 - 9 = +27$. True total = $500 + 27 = 527$, mean = $527 \div 10 = 52.7$. The corrections oppose but the positive one dominates, so the mean rises slightly. Leaving it at 50 ignores the corrections; 54.5 mishandles a sign; 47.3 applies the net correction in the wrong direction.

34. (A) Median first: only positions 1 (smallest) and 7 (largest) changed, so the 4th (middle) value is untouched → median stays 24. Mean: smallest 10 → 2 (a fall of 8) and largest 40 → 60 (a rise of 20), net change +12 to the total, so the mean rises by $12 \div 7 \approx 1.7$ to about 25.7. So the median is fixed while the mean rises. 'Both rise' ignores the median's robustness; 'both stay 24' ignores the unbalanced net change; 'median rises' is false since the middle value never moved.

35. (B) First seven numbers total = $3 \times 14 + 4 \times 21 = 42 + 84 = 126$. Including x makes eight numbers with mean 20, so total = $8 \times 20 = 160$, giving $x = 160 - 126 = 34$. The grand total of the seven (126) is the key intermediate. 20 copies the final mean; 18 is the seven-number mean ($126 \div 7$); 126 is the seven-number total, not x .

36. (A) Increasing the largest value by $30 - 18 = 12$ raises the total by 12, so the mean rises by $12 \div 6 = 2$, to 12. The median of six numbers is the average of the 3rd and 4th sorted values; the largest sits at position 6, so moving it further out does not touch positions 3 or 4 — the median stays 9. The point is the median's robustness to a single extreme: 'both increase' ignores it, 'mean stays 10' ignores the changed total, and 'neither changes' ignores both effects.

37. (C) Totals: Music $15 + 9 = 24$; Art $6 + 22 = 28$, so Art is more popular. Gaps: Music $|15 - 9| = 6$; Art $|6 - 22| = 16$, so Art also has the larger gap. Both halves are correct. Saying Music is more popular reverses the totals; pairing Art-popular with Music-bigger-gap mixes up comparisons; the totals are not equal ($24 \neq 28$).

38. (A) Village P = $2.5 \times 20 = 50$ trees; Village Q = $9 \times 8 = 72$ trees. $Q - P = 72 - 50 = 22$, so Q planted more by 22. The two displays use different scales, so symbol counts and cm cannot be compared directly. 'P is larger' reverses the result; 'same' ignores the conversion; 6.5 wrongly compares 9 against 2.5 without applying either scale.

39. (D) Mean: the total changes by $+6 +6 -6 -6 = 0$, so the mean stays 20. Median: only the two largest (positions 4, 5) and two smallest (positions 1, 2) changed, leaving the 3rd (middle) value untouched, so the median stays 20. Both are unchanged. The balanced ± 6 changes cancel for the mean, and the median is unaffected because the centre never moved; the options claiming a rise or fall ignore one of these facts.

40. (C) The total increase is $1 + 2 + 3 + 4 + 5 = 15$. The original total was $5 \times 12 = 60$, so the new total is $60 + 15 = 75$ over 5 numbers, mean = $75 \div 5 = 15$ (a rise of 3). Leaving the mean at 12 ignores the additions; 17 adds 5 (the last position) to the mean; 13 adds only 1 to the mean. The clean result is 15.

41. (C) Range = largest – smallest = 50. The largest drops by 12 $\rightarrow$ (largest – 12); the smallest rises by 8 $\rightarrow$ (smallest + 8). New range = (largest – 12) – (smallest + 8) = (largest – smallest) – 20 = $50 - 20 = 30$. Both ends move inward, shrinking the spread by $12 + 8 = 20$. Keeping 50 ignores the changes; 70 adds 20 instead of subtracting; 20 subtracts only one shift.

42. (B) Correctly, 30 units need a bar of $30 \div 6 = 5$ cm. The printing error stretches the drawn height by 1.5, so it now measures $5 \times 1.5 = 7.5$ cm, though it still 'means' 30 units. 5 cm is the correct (unstretched) height; 45 cm multiplies units by 1.5 instead of the height; 30 cm confuses units with cm.

43. (D) Total change = $(2 - 6) + (2 - 8) = -4 - 6 = -10$, so the new total drops by 10; the mean falls by $10 \div 6 \approx 1.67$, from 15 to about 13.33 (≈ 13.3) — not exactly 13. For the median of six values (the average of the 3rd and 4th sorted values): the two smallest occupy positions 1 and 2, so lowering them does not touch positions 3 and 4 — the median stays 14. So the mean falls while the median holds at 14. 'Mean stays 15' ignores the

lowered total; 'both fall by 2' wrongly moves the median; 'median also falls to 12' is false because positions 3 and 4 never changed.

44. (C) Initially 7 appears 5 times and 4 appears 3 times. Adding two 4s makes 4 appear $3 + 2 = 5$ times, tying 7's 5. So both 7 and 4 are most frequent — the data is bimodal. Choosing 7 alone ignores that 4 caught up; choosing 4 alone ignores that 7 still has 5; a tie still means there are modes, so 'no mode' is wrong.

45. (C) Let Q have q numbers and P have $2q$. Total = $2q \times 30 + q \times 60 = 60q + 60q = 120q$ over $3q$ numbers, mean = $120q \div 3q = 40$. Because the lower-mean group P is twice as large, the combined mean leans toward 30, landing at 40. The naive average of 30 and 60 is 45 (ignores the sizes); 50 over-weights Q; 90 adds the means.

46. (D) The range $28 - 4 = 24$ describes only the distance between the smallest and largest values. The other three numbers (which sum to $5 \times 14 - 4 - 28 = 70 - 32 = 38$) could be tightly bunched near 14, e.g. 12, 13, 13. So a large range does not force the middle values away from the mean, nor does it pin the median. The range is not the mean, and it makes no claim about every value's distance from 14.

47. (A) Graph distances: $4 \times 5 = 20$, $6 \times 5 = 30$, $8 \times 5 = 40$ km; graph total = 90 km. The odometer reads 95 km, so the graph falls short by $95 - 90 = 5$ km. Adding the units ($4 + 6 + 8 = 18$) without the scale gives 18; 0 assumes agreement; 90 is the graph total, not the shortfall.

48. (C) Sort (already sorted). Mode = 14 (three times). Median = 5th value = 14. Mean = $(8+12+12+14+14+14+16+16+20) \div 9 = 126 \div 9 = 14$. So mean + mode - $2 \times$ median = $14 + 14 - 28 = 0$. Here mean, median and mode all equal 14, making the expression zero. Taking the median as the 4th or 6th value (mis-locating it) gives ± 2 ; 14 is just the common value, not the combined expression.

49. (A) Sorted: 12, 15, 15, 18, 21, 21, 21. Mode = 21 (three times). Median = 4th value = 18. Mean = $(12+15+15+18+21+21+21) \div 7 = 123 \div 7 \approx 17.6$. So 17.6 (mean) < 18 (median) < 21 (mode): mean $<$ median $<$ mode. The single low 12 pulls the mean just below the median, while the cluster of high modal 21s lifts the mode highest. The other orderings misplace the mean relative to the median or wrongly equate the three.

50. (A) The transform new = (old $\div$ 2) + 10 applies to the mean: new mean = $(25 \div 2) + 10 = 12.5 + 10 = 22.5$. Halving alone gives 12.5 (forgets the +10); adding 10 to the original mean before halving, or adding 10 then halving wrong, gives 35; 17.5 mis-halves. The correct new mean is 22.5.

51. (D) Recorded total = $12 \times 40 = 480$. The entry was $91 - 19 = 72$ too low, so the true total = $480 + 72 = 552$, mean = $552 \div 12 = 46$. The error of 72 spread over 12 values raises the mean by 6, to 46. Leaving it at 40 ignores the correction; 43 spreads only half the error; 52 adds 12 (the count) wrongly instead of 6.

52. (D) For nine sorted values the median is the 5th value. The changes touch only positions 1–4 and 6–9; position 5 is never altered and the order is preserved, so the median stays exactly 40. The symmetric ± 100 changes spread the data out far more, but the centre value does not move. Adding 100 to the median (giving 140) or subtracting 100 (giving -60) wrongly assumes the 5th value was among those changed; halving to 20 has no basis. The median's resistance to changes in the tails is the whole point.

53. (A) After 10 innings his total is $10 \times 45 = 450$. To have a mean of 50 over 11 innings, his total must be $11 \times 50 = 550$. So the 11th innings must add $550 - 450 = 100$ runs. Scoring just 50 (the target mean) would give a total of 500 over 11 = 45.5, not 50. The value 55 confuses the score with mean + half the rise; 5 is merely the desired change in the mean.

54. (B) Total of all six = $6 \times 20 = 120$. Three numbers each equal to the mean contribute $3 \times 20 = 60$. So the remaining three sum to $120 - 60 = 60$. The whole total 120 is for all six, not three; 20 is one value; 40 forgets one of the three mean-valued numbers. The other three sum to 60.

55. (D) Basket A = $4 \times 15 = 60$ apples; Basket B = $5 \times 10 = 50$ apples. So A has more, by $60 - 50 = 10$. Even though B shows more SYMBOLS (5 vs 4), the larger scale on graph 1 makes A bigger — symbol counts cannot be compared across different scales. Comparing symbols directly (5 vs 4) wrongly favours B by 1 or claims A by 1; they are not equal.

56. (D) Total = $5 \times 30 = 150$. Subtracting the smallest and largest: $150 - 10 - 60 = 80$ is shared equally by the three middle numbers, so each = $80 \div 3 \approx 26.7$. The value 80 is their combined total, not each one; 30 just copies the overall mean; 40 wrongly divides 80 by 2. Each middle number is about 26.7.

57. (C) Both cities total 180 mm, so seasonal totals match — but matching totals do not mean matching months: P's values are 60, 90, 30 while Q's are 50, 50, 80, a clearly different month-by-month pattern. So 'identical' is wrong even though totals agree. The totals do NOT differ (both 180); equal totals do not make the data identical; and a single largest month (Q's 80) does not mean Q had more rain overall since totals tie.

58. (D) Original total = $7 \times 18 = 126$. New mean 16 over 7 numbers needs a total of $7 \times 16 = 112$, a drop of 14. Replacing 30 by v changes the total by $(v - 30)$, so $v - 30 = -14$, giving $v = 16$. The drop of 14 is the change in total, not the value; 2 is the change in the mean ($18 - 16$); 30 is the original value. The replacement is 16.

59. (C) Adding 100 (well above 40) raises the total, so the mean rises above 40. The mode is the most frequent value; one new 100 cannot outnumber the many 40s, so the mode stays 40. The median, being position-based, shifts only slightly toward the upper-middle (it cannot leap to 100, an extreme). So the mean is most affected, the mode is unmoved, and the median barely moves — the classic outlier pattern. The mode does not become 100

(frequency, not size, defines it); the three do not move together; and adding an extreme value certainly changes the mean.

60. (C) Six numbers total $6 \times 12 = 72$. Eight numbers total $8 \times 15 = 120$, so the 7th and 8th together = $120 - 72 = 48$. Let the 8th be x ; the 7th is $2x$, so $2x + x = 48$, giving $3x = 48$, $x = 16$. The 8th number is 16 (and the 7th is 32). The value 48 is their combined total, not the 8th alone; 24 halves 48 (ignoring that the parts are 2:1, not 1:1); 8 mistakenly splits 48 by 6.